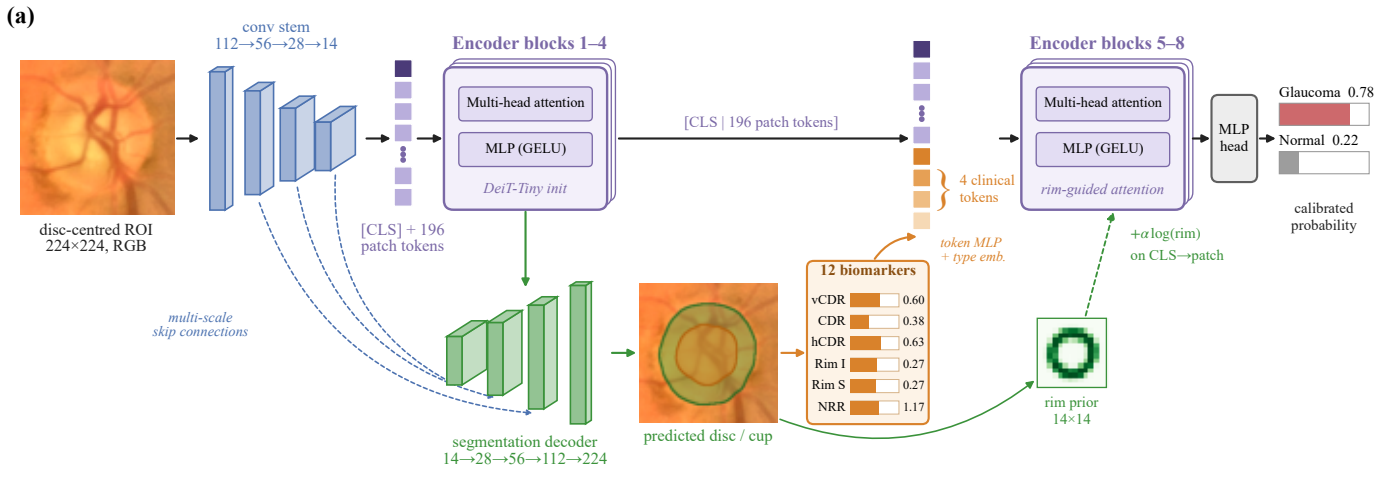
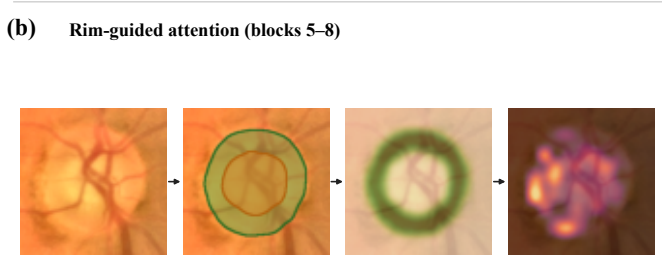
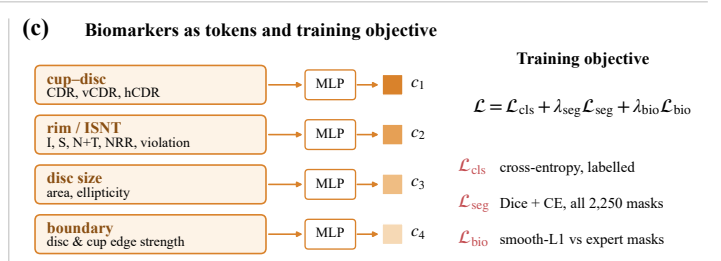

(a)
conv stem
112→56→28→14
Encoder blocks 1–4
Multi-head attention
MLP (GELU)
DeiT-Tiny init
[CLS | 196 patch tokens]
Encoder blocks 5–8
Multi-head attention
MLP (GELU)
rim-guided attention
MLP head
Glaucoma 0.78
Normal 0.22
calibrated probability
disc-centred ROI
224×224, RGB
[CLS] + 196
patch tokens
4 clinical
tokens
token MLP
+ type emb.
+α log(rim)
on CLS→patch
12 biomarkers
vCDR 0.60
CDR 0.38
hCDR 0.63
Rim I 0.27
Rim S 0.27
NRR 1.17
multi-scale
skip connections
segmentation decoder
14→28→56→112→224
predicted disc / cup
rim prior
14×14


(b)
Rim-guided attention (blocks 5–8)
input ROI
predicted disc / cup
rim prior
CLS attention, block 8
$\mathbf{A}_{\mathrm{CLS},j} = \mathrm{softmax}_j\left(\frac{q_{\mathrm{CLS}} k_j^{\top}}{\sqrt{d}} + \alpha_l \log(r_j + \varepsilon)\right)$
r_j: rim probability pooled to the 14×14 patch grid; α_l learned per block


(c)
Biomarkers as tokens and training objective
cup–disc
CDR, vCDR, hCDR
rim / ISNT
I, S, N+T, NRR, violation
disc size
area, ellipticity
boundary
disc & cup edge strength
MLP
c_1
c_2
c_3
c_4
Training objective
$\mathcal{L} = \mathcal{L}_{\mathrm{cls}} + \lambda_{\mathrm{seg}}\mathcal{L}_{\mathrm{seg}} + \lambda_{\mathrm{bio}}\mathcal{L}_{\mathrm{bio}}$
$\mathcal{L}_{\mathrm{cls}}$ cross-entropy, labelled
$\mathcal{L}_{\mathrm{seg}}$ Dice + CE, all 2,250 masks
$\mathcal{L}_{\mathrm{bio}}$ smooth-L1 vs expert masks
$\lambda_{\mathrm{seg}} = 1,\ \lambda_{\mathrm{bio}} = 0.5$
4.91 M parameters · trains on a 4 GB GPU